

HW 2



Prob 3.

$$\begin{aligned} \text{(a)} \quad \hat{y} = Hy &= X(X^T X)^{-1} X^T y, \text{ where } X = UDV^T \\ &= (UDV^T)(UD^T U^T UDV^T)^{-1} U D^T U^T y \\ &= \cancel{UDV^T U^T}^{-1} \cancel{U^{-1}} \cancel{U^T}^{-1} \cancel{U^{-1}} \cancel{V^{-1}} \cancel{V^T} U^T y \\ &= U U^T y \end{aligned}$$

$$\begin{aligned} \text{(b)} \quad \hat{y}_{\text{ridge}} &= X(X^T X + \lambda I)^{-1} X^T y, \quad X = UDV^T \\ &= (UDV^T)(UD^T U^T UDV^T + \lambda I)^{-1} X^T y, \quad Z = UV^T = UU^T \\ &= (UDV^T)(UD^T U^T UDV^T + \lambda UV^T)^{-1} X^T y \\ &= (UDV^T) U (U D^2 + \lambda I) V^T)^{-1} U D^T U^T y \\ &= \cancel{UDV^T U^T}^{-1} \cancel{U^{-1}} \cancel{U^T}^{-1} \cancel{U^{-1}} \cancel{V^{-1}} \cancel{V^T} U^T y \\ &= U D (D^2 + \lambda I)^{-1} D^T U^T y \end{aligned}$$

$$\begin{aligned} \text{(c)} \quad \hat{y}_{\text{ridge}} &= U D \underbrace{(D^2 + \lambda I)^{-1} D^T}_{A \text{ is diagonal}} U^T y. \\ &\quad \text{PxP, } A_{ii} = \frac{d_i^2}{d_i^2 + \lambda}. \\ &= U A U^T y \\ &= \sum_{i=1}^p A_{ii} U_i U_i^T y \\ &= \sum_{i=1}^p \frac{d_i^2}{d_i^2 + \lambda} \cdot U_i U_i^T y. \end{aligned}$$

$$\therefore \lambda = 0, \quad \hat{y}_{\text{ridge}} = \sum_{i=1}^p U_i U_i^T y = U U^T y, \rightarrow \text{linear regression}$$

$\lambda > 0$, $\hat{y}_{\text{ridge}}$ decreases as λ increases,

the smaller the d_i , the larger the shrink effect of x_i .

$\therefore$ shrinks in the direction of small variance.

[illegible]